Lecture 2: Analytical Methods

Introduction to Data Analysis Techniques

1 Lagrange Interpolation

Definition 1.1 (Interpolation). [def:interpolation] Methods to find a smooth function $f(x)$ that passes through a given set of points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$

Theorem 1.1 (Unique Existence of Lagrange Interpolation). [thm:lagrange-existence-uniqueness] *For any set of $n+1$ distinct points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$, there exists a unique polynomial of degree at most n that passes through all these points.*

Convention 1.1 (Lagrange Interpolation Method). [conv:lagrange-interpolation-method] Given a set of $n+1$ distinct points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$. We may construct the following system of equations:

$$\begin{cases} a_0 + a_1 x_0 + a_2 x_0^2 + \ldots + a_n x_0^n &= y_0 \\ a_0 + a_1 x_1 + a_2 x_1^2 + \ldots + a_n x_1^n &= y_1 \\ \vdots & \vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \ldots + a_n x_n^n &= y_n \end{cases} \implies \begin{bmatrix} 1 & x_0 & x_0^2 & \ldots & x_0^n \\ 1 & x_1 & x_1^2 & \ldots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \ldots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

According to the Theorem in Linear Algebra , which states that a system of n linear equations with n unknowns has a unique solution if $\det A \neq 0$, where A is the coefficient matrix.
Since we ask all points in the set to be distinct, we can know that $\det A \neq 0$. Therefore, there exists a unique solution $\vec{a}$
So, we can find a unique polynomial $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n$ that passes through all the given points.

Definition 1.2 (Lagrange Polynomials). [def:lagrange-polynomials]

Example 1.1 (2 Fixed Point). [eg:2-fixed-point] If we are given two points $(x_0, y_0), (x_1, y_1)$,

2 Linear Least Squares Fit

3 General Least Square Fit